

Location-invariant tests of homogeneity of large-dimensional covariance matrices

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ABSTRACT

A test statistic for homogeneity of two or more covariance matrices of large dimensions is presented when the data are multivariate normal. The statistic is location-invariant and defined as a function of U -statistics of non-degenerate kernels so that the corresponding asymptotic theory is employed to derive the limiting normal distribution of the test under a few mild and practical assumptions. Accuracy of the test is shown through simulations with different parameter settings.

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1. Introduction

This article concerns test statistics for homogeneity of several covariance matrices of independent multivariate normal populations with very large dimensions, possibly larger than the sample size. Precisely, let $X_{ik} = (X_{i1k}, \dots, X_{ipk})'$, $k = 1, \dots, n_i$, be n_i iid random vectors with $X_{ik} \sim \mathcal{N}(\mu_i, \Sigma_i)$, $i = 1, \dots, g$, where $\mu_i \in \mathbb{R}^p$, $\Sigma_i \in \mathbb{R}_{>0}^{p \times p}$, represent the usual parameters. We are interested in testing the hypothesis of homogeneity of g covariance matrices, that is,

$$H_0 : \Sigma_1 = \dots = \Sigma_g = \Sigma,$$

when the dimension p may exceed the sample size n_i . For the classical case of $p < n_i$, the likelihood-ratio tests are mostly used to test H_0 . Alternatively, the union–intersection approach may also be used. For details, see for example, Muirhead (2005, Chap. 8) and Seber (2004, Chap. 3). These tests, however, collapse or are inefficient when $p > n_i$ and need to be modified.

The test statistic we propose for H_0 is constructed as a function of unbiased and consistent estimators defined as U -statistics and the corresponding asymptotic theory is used to derive the limiting distribution of the test statistic for $n_i, p \rightarrow \infty$. The test statistic is presented in the next section, beginning with the two-sample case followed by an extension to the general case of g samples. The accuracy of the test through simulations is shown in section 4. Section 5 summarizes the main results. More technical details are collected in the appendix.

2. Test statistic and its properties

2.1. The two-sample case

We begin with the two-sample case, which will help us explain the main concepts and notations easily. An extension to the general case will then be straightforward. Let $\mathbf{X}_{ik} = (X_{ik1}, \dots, X_{ikp})' \sim \mathcal{N}_p(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$, $k = 1, \dots, n_i$, be iid vectors with $\bar{\mathbf{X}}_i = \sum_{k=1}^{n_i} \mathbf{X}_{ik}/n_i$ and $\hat{\boldsymbol{\Sigma}}_i = \sum_{k=1}^{n_i} (\mathbf{X}_{ik} - \bar{\mathbf{X}}_i)(\mathbf{X}_{ik} - \bar{\mathbf{X}}_i)' / (n_i - 1)$ the usual unbiased estimators of $\boldsymbol{\mu}_i$ and $\boldsymbol{\Sigma}_i$, $i = 1, 2$. To test $H_0 : \boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2$, consider the squared Frobenius norm of the difference between the two $\boldsymbol{\Sigma}$'s,

$$\tau_2 = \|\boldsymbol{\Sigma}_1 - \boldsymbol{\Sigma}_2\|_F^2 = \text{tr}(\boldsymbol{\Sigma}_1 - \boldsymbol{\Sigma}_2)^2 = \text{tr}(\boldsymbol{\Sigma}_1^2) + \text{tr}(\boldsymbol{\Sigma}_2^2) - 2\text{tr}(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2),$$

or, in terms of Hilbert space setting, the Hilbert–Schmidt norm. Under normality, Schott (2007) constructs a test for the g -sample case using the same norm with $\hat{\boldsymbol{\Sigma}}_i$ as plug-in estimators for $\boldsymbol{\Sigma}_i$. In our construction of a test for H_0 , say $\hat{\tau}_2$, we define unbiased and consistent estimators of the traces $\text{tr}(\boldsymbol{\Sigma}_1^2)$, $\text{tr}(\boldsymbol{\Sigma}_2^2)$, and $\text{tr}(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2)$, so that $\hat{\tau}_2$ is also an unbiased and consistent estimator of τ_2 .

Note, for example, that $\hat{\boldsymbol{\Sigma}}_i$ and hence $\text{tr}(\hat{\boldsymbol{\Sigma}}_i)$ are unbiased estimator of $\boldsymbol{\Sigma}_i$ and $\text{tr}(\boldsymbol{\Sigma}_i)$, but $\text{tr}(\hat{\boldsymbol{\Sigma}}_i^2)$ is not an unbiased estimator of $\text{tr}(\boldsymbol{\Sigma}_i^2)$. By manipulating certain functions of $\hat{\boldsymbol{\Sigma}}_i$, we define the trace estimators as U -statistics so that they are unbiased and optimal (Polansky 2011). Let E_1, E_2 , and E_{12} be such estimators of the traces so that $\hat{\tau}_2 = E_1 + E_2 - E_{12}$. Write

$$\hat{\tau}_2 - \tau_2 = [E_1 - \text{tr}(\boldsymbol{\Sigma}_1^2)] + [E_2 - \text{tr}(\boldsymbol{\Sigma}_2^2)] - [E_{12} - \text{tr}(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2)], \quad (1)$$

with $E(\hat{\tau}_2 - \tau_2) = 0$. To define E_1, E_2, E_{12} , write $\hat{\boldsymbol{\Sigma}}_i = \sum_{k \neq r}^{n_i} \mathbf{D}_{ikr} \mathbf{D}'_{ikr} / 2Q(n_i)$, $Q(n_i) = n_i(n_i - 1)$, where $\mathbf{D}_{ikr} = \mathbf{X}_{ik} - \mathbf{X}_{ir}$, $k \neq r$, are within-sample differences with $E(\mathbf{D}_{ikr}) = 0$, $\text{Cov}(\mathbf{D}_{ikr}) = 2\boldsymbol{\Sigma}_i$. As $E(\hat{\boldsymbol{\Sigma}}_i) = \boldsymbol{\Sigma}_i \Rightarrow E[\text{tr}(\hat{\boldsymbol{\Sigma}}_i)] = \text{tr}(\boldsymbol{\Sigma}_i)$, $i = 1, 2$, an unbiased estimator of $\text{tr}(\boldsymbol{\Sigma}_i)$, with $A_{ikr} = \mathbf{D}'_{ikr} \mathbf{D}_{1ir}$, is defined as

$$B_i = \frac{1}{Q(n_i)} \sum_{k=1}^{n_i} \sum_{\substack{r=1 \\ k \neq r}}^{n_i} \frac{1}{2} A_{ikr},$$

which is a U -statistic with symmetric kernel $h(\mathbf{X}_{ik}, \mathbf{X}_{ir}) = \mathbf{D}'_{ik} \mathbf{D}_{ir} / 2$. It will help us define similar estimators of $\text{tr}(\boldsymbol{\Sigma}_i^2)$ and $\text{tr}(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2)$. Note that $E(\mathbf{D}_{ikr} \mathbf{D}'_{ikr} \mathbf{D}_{ik'r'} \mathbf{D}'_{ik'r'}) = 4\boldsymbol{\Sigma}_i^2$ by independence, if $k \neq k'$, $r \neq r'$, $k \neq r'$, $k' \neq r$. Let $A_{ikrk'r'}^2 = (\mathbf{D}'_{ikr} \mathbf{D}_{ik'r'})^2$; similarly, $A_{ikk'r'r'}^2$, $A_{ikr'r'k'}^2$. Then, with $D_{ikrk'r'}^2 = A_{ikrk'r'}^2 + A_{ikk'r'r'}^2 + A_{ikr'r'k'}^2$, $P(n_i) = n_i(n_i - 1)(n_i - 2)(n_i - 3)$,

$$E_i = \frac{1}{12P(n_i)} \sum_{k=1}^{n_i} \sum_{r=1}^{n_i} \sum_{\substack{k'=1 \\ k \neq k'}}^{n_i} \sum_{\substack{r'=1 \\ r \neq r'}}^{n_i} D_{ikrk'r'}^2 \quad (2)$$

is an unbiased estimator of $\text{tr}(\boldsymbol{\Sigma}_i^2)$ where $\pi(\cdot)$ denotes all indices unequal. E_i is a U -statistic with symmetric kernel $h(\mathbf{X}_{ik}, \mathbf{X}_{ir}, \mathbf{X}_{ik'}, \mathbf{X}_{ir'}) = D_{ikrk'r'}^2 / 4$, where $E(D_{ikrk'r'}^2) = 3[4\{\text{tr}(\boldsymbol{\Sigma}_i^2)\}]$. As

$E(\hat{\Sigma}_1 \hat{\Sigma}_2) = \Sigma_1 \Sigma_2 \Rightarrow E[\text{tr}(\hat{\Sigma}_1 \hat{\Sigma}_2)] = \text{tr}(\Sigma_1 \Sigma_2)$ by the independence of samples, an unbiased estimator of $\text{tr}(\Sigma_1 \Sigma_2)$ is given as

$$E_{12} = \frac{1}{4P(n_1, n_2)} \sum_{\substack{k=1 \\ k \neq r}}^{n_1} \sum_{r=1}^{n_1} \sum_{\substack{l=1 \\ l \neq s}}^{n_2} \sum_{s=1}^{n_2} A_{12klrs}^2, \quad (3)$$

where $A_{12klrs}^2 = (\mathbf{D}'_{1kr} \mathbf{D}_{2ls})^2$ and $P(n_1, n_2) = Q(n_1)Q(n_2)$. That E_i and E_{12} are not only unbiased but also consistent estimators, even when $p \gg n_i$, is ensured by Theorem 2.1.1, proved in the appendix, section B.1. As stated already, there are many ways to construct unbiased estimators of these traces. For example, Ahmad (2015) gives unbiased and consistent estimators of the three traces while presenting a statistic for two-sample mean testing problem. The estimators are, however, not location-invariant so that $\mu_i = 0$ needs to be assumed. As E_i, E_{12} are location-invariant, no such assumption is required.

Further, defining E_i and E_{12} as a U -statistics class of estimators helps us use corresponding theory to study the properties of the proposed test. In this context, note that an unbiased E_i can also be defined using only one of the components in its kernel, that is, $h(\mathbf{X}_{ik}, \mathbf{X}_{ir}, \mathbf{X}_{ik'}, \mathbf{X}_{ir'}) = A_{ikrk'r'}^2$, but now $h(\cdot)$ is not symmetric and the theory of U -statistics with symmetric kernels cannot be applied. Finally, the kernels of all estimators are the squared pairwise Euclidean distances between vectors, that is, $\|\mathbf{x}_{ik} - \mathbf{x}_{ir}\|^2$, and so on. Estimators of this type are frequently used in classical and modern multivariate theory; see, for examples, Seber (2004, 277, 394) or Gretton et al. (2008). Now, to proceed further with T_2 , we begin with the moment properties of estimators in the following theorem.

Theorem 2.1.1. *Let E_i and E_{12} be as defined in Eqs. (2) and (3), respectively. Then $E(E_i) = \text{tr}(\Sigma_i^2)$, $E(E_{12}) = \text{tr}(\Sigma_1 \Sigma_2)$, and*

$$\begin{aligned} \text{Var}(E_i) &= \frac{1}{P(n_i)} [(2n_i^3 - 9n_i^2 + 9n_i - 16)\text{tr}(\Sigma_i^4) + (n_i^2 - 3n_i + 8)[\text{tr}(\Sigma_i^2)]^2] \\ \text{Var}(E_{12}) &= \frac{2}{(n_1 - 1)(n_2 - 1)} [(n_1 + n_2 - 1)\text{tr}\{(\Sigma_1 \Sigma_2)^2\} + [\text{tr}(\Sigma_1 \Sigma_2)]^2] \\ \text{Cov}(E_i, E_{12}) &= \frac{4}{n_i - 1} \text{tr}(\Sigma_i^3 \Sigma_j), i \neq j = 1, 2; \text{Cov}(E_1, E_2) = 0. \end{aligned}$$

Further, $\text{Var}(E_i/\text{tr}(\Sigma_i^2)) = O(1/n_i)$, $\text{Var}(E_{12}/\text{tr}(\Sigma_1 \Sigma_2)) = O(1/n_1 + 1/n_2)$, and $\text{Cov}(E_i/\text{tr}(\Sigma_i^2), E_{12}/\text{tr}(\Sigma_1 \Sigma_2)) = O(1/n_i)$.

By Theorem 2.1.1 ratios of the moments of estimators to their traces are uniformly bounded and independent of p . This fact will help us derive the non-degenerate limit of T_2 for $n_i, p \rightarrow \infty$. To see this, write $T_2 = (\hat{\tau}_2 - \tau_2)/p^2$ such that

$$T_2 = \sum_{i=1}^2 a_i \tilde{E}_i - 2a_{12} \tilde{E}_{12}, \quad (4)$$

with $a_i = \text{tr}(\Sigma_i^2)/p^2$, $a_{12} = \text{tr}(\Sigma_1 \Sigma_2)/p^2$, and $\tilde{E} = E/\text{tr}(E) - 1$ with $E(\tilde{E}) = 0$, so that Theorem 2.1.1 essentially deals with the properties of $\tilde{E}$. To take care of the behavior of a_i and a_{12} , we set certain assumptions. For $n_i, p \rightarrow \infty$, let:

Assumption 2.1.1. $\text{tr}(\Sigma)/p = O(1)$.

Assumption 2.1.2. (i) $p/n_i \rightarrow c_i \leq c \in (0, \infty)$ (ii) $n_1/n_2 \rightarrow c_3 \in (0, \infty)$.

As an immediate consequence of Assumption 2.1.1, $a = \text{tr}(\Sigma^2)/p^2 = O(1)$, so that a reference to Assumption 2.1.1 in the sequel may also refer to its consequence, where a denotes a_i under H_0 . If we let λ_j denote the eigenvalues of Σ so that $\nu_j = \lambda_j/p$ are the eigenvalues of Σ/p , then Assumption 2.1.1 and its consequence bound the first two moments of ν_j .

These assumptions hold for a wide variety of covariance structures. Consider, for example, the compound symmetric structure $\Sigma = (1 - \rho)\mathbf{I} + \rho\mathbf{J}$ with $\mathbf{I}$ as identity matrix, $\mathbf{J} = \mathbf{1}\mathbf{1}'$, $\mathbf{1}$ a vectors of 1s, and $\rho \in \mathbb{R}$, $-1/(p-1) \leq \rho \leq 1$ (section 4). Note that $\text{tr}(\Sigma^i) = O(p^i)$, $i = 1, 2$, satisfying the assumptions.

The CS structure typically belongs to the class of spiked covariance matrices where a few eigenvalues dominate the rest. There is an increasing interest in high-dimensional inference for spiked covariance matrices. For a few recent references, see Jung and Marron (2009), Yata and Aoshima (2012; 2013), and Aoshima and Yata (2016). On the other hand, AR(1) and UN structures, also used in section 4, are examples of nonspiked cases. Two particular nonspiked cases are when common Σ has spherical or identity structure, $\Sigma \propto I$ or $\Sigma = I$. As they belong to a very special class of hypotheses in multivariate theory, they are not considered here. Ahmad (2017) provides a detailed treatment of multi-sample tests of the two hypotheses, assuming H_0 of homogeneity holds.

We need one more result to be able to state the main theorem. To the unbiasedness and consistency properties of the estimators in Theorem 2.1.1, the following theorem, proved in Appendix section B.2, adds asymptotic normality.

Theorem 2.1.2. *Let $\tilde{E}_i$ and $\tilde{E}_{12}$ be as already stated. Then, as $n_i, p \rightarrow \infty$*

$$\tilde{E}_i/\sqrt{\text{Var}(\tilde{E}_i)} \xrightarrow{\mathcal{D}} N(0, 1) \text{ and } \tilde{E}_{12}/\sqrt{\text{Var}(\tilde{E}_{12})} \xrightarrow{\mathcal{D}} N(0, 1).$$

To realize the essence of Theorem 2.1.2., recall that E_i and E_{12} are U -statistics so that $\tilde{E} = E/E(E) - 1$ are their standardized forms with kernels $\tilde{h}(\cdot) = h/E[h(\cdot)] - 1$ with $E[h(\cdot)] = E(U) = E(E)$ (see Appendix section A.1). Then the proof follows from the asymptotic theory of U -statistics by taking care of arbitrary p . The main requirement is of square-integrability of $\tilde{h}(\cdot)$, which is ensured by $a = O(1)$. The details are left to Appendix section B.2. For more on the general theory, see, for example; Serfling (1980), van der Vaart (1998), and Koroljuk and Borovskich (1994). Now, Theorems 2.1.1 and 2.1.2 imply that T_2 is a linear combination of unbiased, consistent and asymptotically normal U -statistics with $E(T_2) = 0$ and

$$\sigma_{T_2}^2 = \sum_{i=1}^2 a_i^2 \text{Var}(\tilde{E}_i) + 4a_{12}^2 \text{Var}(\tilde{E}_{12}) - 4a_{12} \sum_{i=1}^2 a_i \text{Cov}(\tilde{E}_i, \tilde{E}_{12}), \quad (5)$$

where $\sigma_{T_2}^2 = \text{Var}(T_2)$. Under $H_0 : \Sigma_1 = \Sigma_2 = \Sigma$, T_2 and $\text{Var}(T_2)$ simplify to

$$T_2 = a[(E_1 + E_2 - 2E_{12})/\text{tr}(\Sigma^2)] \quad (6)$$

$$\sigma_{T_2}^2 = a^2[O(n_0)\kappa + 4n_0^2] \quad (7)$$

with $n_0 = n/n_1n_2$, $n = n_1 + n_2$, where $\kappa = \text{tr}(\Sigma^4)/[\text{tr}(\Sigma^2)]^2$. Under Assumption 2.1.2, the first part of $\sigma_{T_2}^2$ vanishes for $n_i, p \rightarrow \infty$. Ignoring these vanishing terms, $\sigma_{T_2}^2$ under H_0 can be approximated as

$$\sigma_{T_{20}}^2 = 4a^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)^2. \quad (8)$$

With T_2 as a linear combination of asymptotically normal components, two of which are independent and their covariances with the third asymptotically vanish, the asymptotic normality of T_2 , as $n_i, p \rightarrow \infty$, follows by a simple application of the Cramér–Wold device (van der Vaart 1998, 16). We state the result without proof. For a detailed proof in a similar setting, see for example, Jiang (2010, chap. 11) or Koroljuk and Borovskich (1994, chap. 4).

Theorem 2.1.3. For T_2 in Eq. (4), $\sigma_{T_2}^{-1}(T_2 - \tau_2) \xrightarrow{\mathcal{D}} N(0, 1)$ under Assumptions 2.1.1 and 2.1.2, as $n_i, p \rightarrow \infty$ with $\sigma_{T_2}^2$ given in Eq. (5). Further, $\sigma_{T_{20}}^{-1}T_2 \xrightarrow{\mathcal{D}} N(0, 1)$ under H_0 with $\sigma_{T_{20}}^2$ given in Eq. (8).

Let z_α denote the $100\alpha\%$ th quantile of $N(0, 1)$, $\beta(\boldsymbol{\theta})$ be the power function of T_2 with $\boldsymbol{\theta} = \{\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2\}$, and Θ_0, Θ_1 be the parameter spaces under H_0 and H_1 , where $\Theta_0 \cup \Theta_1 = \boldsymbol{\theta}$, $\Theta_0 \cap \Theta_1 = \emptyset$. Let Assumption 2.1.1 hold for each population, that is, $\text{tr}(\boldsymbol{\Sigma}_i)/p = O(1)$, $i = 1, 2$. By Theorem 2.1.3, $\beta(\boldsymbol{\theta}) = P((T_2 - \tau_2)/\sigma_{T_{20}} \geq z_\alpha)$ with $\beta(\boldsymbol{\theta}|H_0) = \alpha$, $\beta(\boldsymbol{\theta}|H_1) = 1 - \beta$, so that $\beta(\boldsymbol{\theta}|\boldsymbol{\theta} \in \Theta) = P(T_2/\sigma_{T_2} \geq \gamma z_\alpha + \delta)$, where $\gamma = \sigma_{T_{20}}/\sigma_{T_2}$ and $\delta = \tau/\sigma_{T_2}$. Since $P(T_2/\sigma_{T_2} \geq \gamma z_\alpha + \delta) = 1 - \Phi(\gamma z_\alpha + \delta)$, where $\Phi(\cdot)$ is the cdf of normal distribution, the asymptotic power follows as

$$\lim_{n_i, p \rightarrow \infty} P(T_2/\sigma_{T_2} \geq \gamma z_\alpha + \delta) = \lim_{n_i, p \rightarrow \infty} [1 - \Phi(\gamma z_\alpha + \delta)].$$

Similarly, using Eq. (8) and Theorem 2.1.1, it trivially follows that $\gamma = O(1)$, so that the limiting power simplifies as $\lim_{n_i, p \rightarrow \infty} [1 - \Phi(z_\alpha + \delta)]$, which reduces to α under H_0 . Moreover, since δ simplifies to be of order $O(p)$, the power increases with increasing p . This feature of the test statistic will become more evident through simulation results in section. 4.

To particularly observe the behavior of power function under the local alternative, let $\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}$, $\boldsymbol{\Sigma}_2 = \boldsymbol{\Sigma} + A/\sqrt{n}$, where $A_p \geq 0$. Let $\text{tr}(\mathbf{A}^2)/p^2 = O(1)$ (as a consequence of Assumption 2.1.1 with $\boldsymbol{\Sigma} = \mathbf{A}$). This implies that the same power function holds as computed earlier with $\tau_2 = \|\mathbf{A}\|^2/n$.

For practical application of T_2 , we need to estimate a . Since E_{12} is composed of information from both samples and $E(E_{12}) = \text{tr}(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2) = \text{tr}(\boldsymbol{\Sigma}^2)$ under H_0 , we use $\hat{a} = E_{12}/p^2$ to estimate a . The properties of $\hat{a}$, particularly consistency, follow from Theorems 2.1.1 and 2.1.2. With $\text{Var}(E_{12}) = O(n_0)$, it follows, by replacing a with $\hat{a}$ in $\sigma_{T_2}^2$, that $\hat{\sigma}_{T_2}^2$ is also a consistent estimator such that $\hat{\sigma}_{T_2}/\sigma_{T_2}^2 \rightarrow 1$. In particular, under H_0 , $\hat{\sigma}_{T_{20}}^2 = 4\hat{a}^2 n_0^2$ can be used for $\sigma_{T_{20}}^2$, $n_0 = 1/n_1 + 1/n_2$. The following corollary can now replace Theorem 2.1.3.

Corollary 2.1.1. Theorem 2.1.3 remains valid with a replaced with $\hat{a}$ in $\sigma_{T_2}^2$ or $\sigma_{T_{20}}^2$.

Before moving on to the multisample extension, a remark on the computational efficiency is in order. Let $\bar{\mathbf{X}}_i = \sum_{k=1}^{n_i} \mathbf{X}_{ki}/n_i$, $\hat{\boldsymbol{\Sigma}}_i = \sum_{k=1}^{n_i} \tilde{\mathbf{X}}_{ki}\tilde{\mathbf{X}}_{ki}'/(n_i - 1)$ be estimators of $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$, with $\tilde{\mathbf{X}}_{ki} = \mathbf{X}_{ki} - \bar{\mathbf{X}}_i$. Let $Q_i = \sum_{k=1}^{n_i} (\tilde{\mathbf{X}}_{ki}'\tilde{\mathbf{X}}_{ki})^2/(n_i - 1)$, $\eta_i = (n_i - 1)/[n_i(n_i - 2)(n_i - 3)]$, $i = 1, 2$. Then, we can alternatively write $E_i = \eta_i \left\{ (n_i - 1)(n_i - 2)\text{tr}(\hat{\boldsymbol{\Sigma}}_i^2) + [\text{tr}(\hat{\boldsymbol{\Sigma}}_i)]^2 - n_i Q_i \right\}$ (see, e.g., Himeno and Yamada, 2014), where $E_{12} = \text{tr}(\hat{\boldsymbol{\Sigma}}_1\hat{\boldsymbol{\Sigma}}_2)$. An extension to the multisample case follows similarly. Although the U -statistics forms of

estimators in Eqs. (2) and (3) make it convenient to study the properties of the test statistic, the alternative forms given in the preceding drastically reduce the computational cost. The simulations in [section 4](#) are produced using this alternative form.

3. Multisample extension

With most of the notations and concepts clarified in [section 2.1](#), the extension of T_2 to T_g for $H_0 : \Sigma_i = \Sigma$, $i = 1, \dots, g$, is now straightforward. We therefore restrict this section to only necessary computations and details and mostly refer to the corresponding results for $g = 2$ already described. The estimators

$$E_i = \frac{1}{12P(n_i)} \sum_{k=1}^{n_i} \sum_{r=1}^{n_i} \sum_{k'=1}^{n_i} \sum_{r'=1}^{n_i} D_{ikrk'r'}^2, \quad E_{ij} = \frac{1}{4P(n_i, n_j)} \sum_{k=1}^{n_i} \sum_{r=1}^{n_i} \sum_{l=1}^{n_j} \sum_{s=1}^{n_j} A_{ijkrls}^2, \quad (9)$$

$\pi(k, r, k', r')$ $k \neq r$ $l \neq s$

are the same except for $i, j = 1, \dots, g$, $P(n_i) = n_i(n_i - 1)(n_i - 2)(n_i - 3)$, $P(n_i, n_j) = Q(n_i)Q(n_j)$, with $D_{ikrk'r'}^2 = A_{ikrk'r'}^2 + A_{ikk'r'r'}^2 + A_{ikr'k'l'}^2$, $A_{ikrk'r'}^2 = (\mathbf{D}'_{ikr} \mathbf{D}_{ik'r'})^2$, $A_{ijkrls}^2 = (\mathbf{D}'_{ikr} \mathbf{D}_{jls})^2$, $\mathbf{D}_{ikr} = \mathbf{X}_{ik} - \mathbf{X}_{ir}$, $\mathbf{D}_{jls} = \mathbf{X}_{jl} - \mathbf{X}_{js}$, $i \neq j$, and $\pi(\cdot)$ denotes again all indices pairwise unequal. The test statistic takes the form

$$T_g = \sum_{i=1}^g \sum_{j=1}^g T_{2ij} = (g-1) \sum_{i=1}^g T_i - 2 \sum_{i=1}^g \sum_{j=1}^g T_{ij}, \quad (10)$$

$i < j$ $i < j$

with $T_i = a_i \tilde{E}_i$, $T_{ij} = a_{ij} \tilde{E}_{ij}$, and $a_i = \text{tr}(\Sigma_i^2)/p^2$, and $a_{ij} = \text{tr}(\Sigma_i \Sigma_j)/p^2$, where T_{2ij} is T_2 defined for the (i, j) th pair of independent samples. Thus, T_g extends T_2 directly as a sum of Frobenius norms over all distinct pairs of g samples, that is, $\tau_g = \sum_{i < j} \|\Sigma_i - \Sigma_j\|^2$, scaled by p^2 . Clearly, $E(T_g) = 0$, and

$$\begin{aligned} \sigma_{T_g}^2 &= (g-1)^2 \sum_{i=1}^g \text{Var}(T_i) + 4 \sum_{i=1}^g \sum_{j=1}^g \text{Var}(T_{ij}) + 8 \sum_{i=1}^g \sum_{j=1}^g \sum_{j'=1}^g \text{Cov}(T_{ij}, T_{ij'}) \\ &\quad - 8(g-1) \sum_{i=1}^g \sum_{j=1}^g \text{Cov}(T_i, T_{ij}) - 8(g-1) \sum_{i=1}^g \sum_{j=1}^g \sum_{j'=1}^g \text{Cov}(T_i, T_{ij'}) \\ &\quad + 8 \sum_{i=1}^g \sum_{j=1}^g \sum_{j'=1}^g \text{Cov}(T_{ij}, T_{ij'}) + 8 \sum_{i=1}^g \sum_{j=1}^g \sum_{j'=1}^g \text{Cov}(T_{ij}, T_{ij'}) \end{aligned} \quad (11)$$

$i < j$ $i < j < j'$ $i < j' < j$

Theorem 2.1.1 gives $\text{Var}(T_i)$ and $\text{Var}(T_{ij})$. Covariances with unequal indices vanish. For nonzero covariances, consider $\text{Cov}(T_{ij}, T_{ij'}) = a_{ij} a_{ij'} \text{Cov}(\tilde{E}_{ij}, \tilde{E}_{ij'})$ with

$$\text{Cov}(E_{ij}, E_{ij'}) = \frac{1}{16P(n_i, n_j)P(n_i, n_{j'})} \sum_{k=1}^{n_i} \sum_{r=1}^{n_i} \sum_{l=1}^{n_j} \sum_{s=1}^{n_j} \sum_{l'=1}^{n_{j'}} \sum_{s'=1}^{n_{j'}} \text{Cov}(A_{ijkrls}^2, A_{ij'krls}^2).$$

$k \neq r$ $l \neq s$ $l' \neq s'$

From Theorem A.2.1 and Lemma A.2.1 in the appendix, $\text{Cov}(E_{ij}, E_{ij'}) = 2\text{tr}(\Sigma_i \Sigma_j \Sigma_i \Sigma_{j'}) / Q(n_i)$ since

$$\begin{aligned} E(A_{ijkr}^2 A_{ij'krls}^2) &= E(\mathbf{D}'_{ikr} \mathbf{D}_{jls} \mathbf{D}'_{jls} \mathbf{D}_{ikr} \mathbf{D}'_{ikr} \mathbf{D}_{j'ls} \mathbf{D}_{j'ls} \mathbf{D}_{ikr}) \\ &= 4E(\mathbf{D}'_{ikr} \Sigma_j \mathbf{D}_{ikr} \mathbf{D}'_{ikr} \Sigma_{j'} \mathbf{D}_{ikr}) \\ &= 32\text{tr}(\Sigma_i \Sigma_j \Sigma_i \Sigma_{j'}) + 16\text{tr}(\Sigma_i \Sigma_j) \text{tr}(\Sigma_i \Sigma_{j'}). \end{aligned}$$

Similarly, $\text{Cov}(T_{ij}, T_{ij'})$. As $\text{Cov}(E_i, E_{ij})$ follows from $\text{Cov}(E_1, E_{12})$ in Theorem 2.1.1, it gives $\text{Cov}(T_i, T_{ij})$ and $\text{Cov}(T_i, T_{ji})$. Substituting the results and noting that $\text{Cov}(\tilde{E}_{ij}, \tilde{E}_{ij'}) = \text{Cov}(E_{ij}/\text{tr}(\Sigma_i \Sigma_j), E_{ij'}/\text{tr}(\Sigma_i \Sigma_{j'}))$, we get

$$\begin{aligned} \sigma_{T_g}^2 &= (g-1)^2 \sum_{i=1}^g a_i^2 \text{Var}(\tilde{E}_i) + 4 \sum_{i=1}^g \sum_{j=1}^g a_{ij}^2 \text{Var}(\tilde{E}_{ij}) \\ &\quad + 8 \sum_{i=1}^g \sum_{j=1}^g \sum_{j'=1}^g a_{ij} a_{ij'} \text{Cov}(\tilde{E}_{ij}, \tilde{E}_{ij'}) + 8 \sum_{i=1}^g \sum_{j'=1}^g \sum_{j=1}^g a_{ij} a_{j'j} \text{Cov}(\tilde{E}_{ij}, \tilde{E}_{j'j}) \\ &\quad - 8(g-1) \sum_{i=1}^g \sum_{j=1}^g a_i a_{ij} \text{Cov}(\tilde{E}_i, \tilde{E}_{ij}) \end{aligned} \quad (12)$$

where variances and covariances are as computed earlier. In particular, under H_{0g} ,

$$\begin{aligned} T_g &= a \left(\frac{(g-1) \sum_{i=1}^g E_i - 2 \sum_{i \neq j}^g E_{ij}}{\text{tr}(\Sigma^2)} \right) \\ \sigma_{T_{g0}}^2 &= a^2 \left[O\left(\sum_{i=1}^g \frac{1}{n_i}\right) \kappa + 4 \left((g-1)^2 \sum_{i=1}^g \frac{1}{n_i^2} + \sum_{i=1}^g \sum_{j=1}^g \frac{2}{n_i n_j} \right) \right]. \end{aligned}$$

Following the same arguments as for $g=2$, $\sigma_{T_{g0}}^2$ can be approximated as

$$\sigma_{T_{g0}}^2 = 4a^2 \left[(g-1)^2 \sum_{i=1}^g \frac{1}{n_i^2} + \sum_{i=1}^g \sum_{j=1}^g \frac{2}{n_i n_j} \right], \quad (13)$$

under the assumptions. With $\hat{a} = \sum_{i \neq j}^g P(n_i, n_j) E_{ij} / P^*$, $P^* = \sum_{i \neq j}^g P(n_i, n_j)$ plugged in $\sigma_{T_g}^2$ as a consistent estimator of a , $\hat{\sigma}_{T_g}^2$ can be used as a consistent estimator of $\sigma_{T_g}^2$. All results of section 2.1 extend without any inconvenience. We only summarize Theorem 2.1.3 and Corollary 2.1 in the following for T_g .

Theorem 3.1. For T_g in Eq. (10), $\sigma_{T_g}^{-1}(T_g - \tau_g) \xrightarrow{\mathcal{D}} N(0, 1)$, under Assumptions 2.1.1 and 2.1.2, as $n_i, p \rightarrow \infty$ with $\sigma_{T_g}^2$ in Eq. (13) Under H_0 , $\sigma_{T_{g0}}^{-1} T_g \xrightarrow{\mathcal{D}} N(0, 1)$ with $\sigma_{T_{g0}}^2$ in Eq. (8) Further, the limits remain valid by replacing a with $\hat{a}$ in $\sigma_{T_g}^2$.

4. Simulations

We assess the accuracy of T_g for its size control and power. For $g = 2$, we generate data from two independent $\mathcal{N}_p(0, \Sigma_i)$ distributions with $(n_1, n_2) = (5, 10), (10, 15)$ and $p = \{20, 50, 100, 300, 500, 1000\}$. For Σ_i , we use three structures, compound symmetry (CS), autoregressive of order 1, AR(1), and unstructured (UN), where CS is defined after the assumptions, AR(1) is defined as $\text{Cov}(X_k, X_l) = \kappa\rho^{|k-l|}$, $\forall k, l$, and UN is defined as $\Sigma = (\sigma_{ij})_{i,j=1}^d$, $\sigma_{ij} = 1(i = j)$, $\rho_{ij} = (i - 1)/d$ ($i > j$). We use $\kappa = 1$, $\rho = 0.5$ for CS and AR(1). Under H_1 , we use pairwise covariance matrices for two populations. Under H_0 , we keep it the same for both populations.

For test size, we use $1 - \alpha = 0.90, 0.95, 0.99$ and estimate the nominal value through an average of 1000 simulation runs under each of the aforementioned parameter sets. For power $1 - \beta$, we fix $\alpha = 0.95$ and estimate it similarly under the corresponding parameter sets. Tables 1 and 2 report the results for estimated $1 - \alpha$ and $1 - \beta$, respectively. The test size under UN structure is not reported due to results very similar to those of other structures.

We observe an improving accuracy of T_2 with increasing p , particularly for $p \geq 50$ in all cases. We also note a discernable improvement in size control for only a slight increase in the sample sizes. Similarly, the power also improves drastically with increasing p . The low power values for the CS–AR covariance pair improve reasonably well for sample sizes

Table 1. Estimated $1 - \alpha$ of T_g for $g = 2$.

n_1, n_2	Σ :	CS			AR(1)		
	p	0.90	0.95	0.99	0.90	0.95	0.99
5, 10	20	0.950	0.966	0.985	0.926	0.948	0.983
	50	0.915	0.950	0.981	0.919	0.953	0.987
	100	0.913	0.952	0.980	0.917	0.950	0.980
	300	0.905	0.951	0.988	0.908	0.947	0.989
	500	0.901	0.954	0.991	0.904	0.949	0.983
	1000	0.904	0.955	0.990	0.901	0.953	0.989
10, 15	20	0.924	0.944	0.970	0.911	0.950	0.981
	50	0.922	0.940	0.975	0.898	0.942	0.985
	100	0.912	0.954	0.982	0.891	0.941	0.987
	300	0.898	0.947	0.991	0.910	0.949	0.993
	500	0.910	0.955	0.993	0.907	0.945	0.991
	1000	0.909	0.952	0.990	0.901	0.950	0.988

Table 2. Estimated $1 - \beta$ of T_g for $g = 2$.

n_1, n_2	p	CS-AR	AR-UN	CS-UN
5, 10	20	0.315	0.384	0.623
	50	0.551	0.456	0.799
	100	0.713	0.503	0.892
	300	0.908	0.525	0.989
	500	0.999	0.613	1.000
	1000	1.000	0.709	1.000
10, 15	20	0.598	0.798	0.953
	50	0.920	0.924	1.000
	100	0.998	1.000	1.000
	300	1.000	1.000	1.000
	500	1.000	1.000	1.000
	1000	1.000	1.000	1.000

(10, 15). It can be concluded that the test statistic performs accurately for very small sample sizes and any arbitrary dimension and under a variety of practically useful covariance matrices.

The test was also evaluated for other parameters, for example, $\rho = 0.2$, $\rho = 0.8$ for covariance structures, with results similar to those reported here. Similarly, it was found that the accuracy of the test improves by increasing n_i even if the design is seriously unbalanced, for example, $(n_1, n_2) = (20, 30), (10, 60), (5, 50)$. The results presented here are particularly selected to show the accuracy for small n .

5. Discussion

A test statistic for homogeneity of several covariance matrices coming from multivariate normal populations is presented when the dimension is very large, possibly much larger than the sample size. Such a test is usually carried out in multivariate theory, for examples, as a preliminary test of homoscedasticity in multivariate analysis of variance, or in discriminant analysis, or in model-based clustering, where this test combined with the test of equality of mean vectors is usually referred to as the test of equality of distributions.

The test statistic is formulated as a linear combination of location-invariant estimators defined as U -statistics of higher order non-degenerate kernels. The corresponding asymptotic theory is then used to derive the asymptotic normal limit of the proposed test when both sample size and dimension go to infinity. Simulation results are used to show the accuracy of the test statistic with respect to its size control and power for very large sample sizes, arbitrary dimensions, and different covariance matrices often used in multivariate analysis.

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References

- Ahmad, M. R. 2014. Location-invariant and non-invariant tests for high-dimensional covariance matrices under normality and non-normality. Technical report, Uppsala University, Uppsala, Sweden.
- Ahmad, M. R. 2015. U -tests of general linear hypotheses for high-dimensional data under non-normality and heteroscedasticity. *Journal of Statistical Theory & Practice* 90 (3):544–70.
- Ahmad, M. R. 2017. Location-invariant multi-sample U -tests for covariance matrices with large dimension. *Scandinavian Journal of Statistics*, in press. doi:[10.1111/sjos.12262](https://doi.org/10.1111/sjos.12262).
- Aoshima, M., and K. Yata. 2016. Two-sample tests for high-dimension, strongly spiked eigenvalue models. arXiv 1602.02491.

- Gretton, A., K. M. Borgwardt, M. J. Rasch, et al. 2008. A kernel method for the two-sample problem. *Journal of Machine Learning Research* 1:1–43.
- Himeno, T., and T. Yamada. 2014. Estimations for some functions of covariance matrix in high dimension under non-normality and its applications. *Journal of Multivariate Analysis* 130:27–44.
- Hoeffding, W. 1948. A class of statistics with asymptotically normal distribution. *Annals of Mathematical Statistics* 19:293–325.
- Jiang, J. 2010. *Large sample techniques for statistics*. New York, NY: Springer.
- Jung, S., and J. S. Marron. 2009. PCA consistency in high dimension, low sample size context. *Annals of Statistics* 37:4104–30.
- Koroljuk, V. S., and Y. V. Borovskich. 1994. *Theory of U-statistics*. Dordrecht, The Netherlands: Kluwer Academic Press.
- Lee, A. J. 1990. *U-statistics: Theory and Practice*. Boca Raton, FL: CRC Press.
- Lehmann, E. L. 1999. *Elements of large-sample theory*. New York, NY: Springer.
- Mathai, A. M., S. B. Provost, and T. Hayakawa. 1995. *Bilinear forms and zonal polynomials*. Lecture Notes in Statistics, no. 102. New York, NY: Springer.
- Muirhead, R. J. 2005. *Aspects of multivariate statistical theory*. New York, NY: Wiley.
- Polansky, A. M. 2011. *Introduction to statistical limit theory*. Boca Raton, FL: CRC Press.
- Schott, J. R. 2007. A test for the equality of covariance matrices when the dimension is large relative to the sample size. *Computational Statistics & Data Analysis* 51:6535–6542.
- Searle, S. R. 1971. *Linear models*. New York, NY: Wiley.
- Seber, G. A. F. 2004. *Multivariate observations*. New York, NY: Wiley.
- Serfling, R. J. 1980. *Approximation theorems of mathematical statistics*. Weinheim, Germany: Wiley.
- van der Vaart, A. W. 1998. *Asymptotic statistics*. New York, NY: Cambridge University Press.
- Yata, K., and M. Aoshima. 2012. Effective PCA for high-dimension, low-sample-size data with noise reduction via geometric representations. *Journal of Multivariate Analysis* 105:193–215.
- Yata, K., and M. Aoshima. 2013. PCA consistency for the power spiked model in high-dimensional settings. *Journal of Multivariate Analysis* 122:334–354.

Appendix A: Basic results

A.1. U-statistics

We briefly summarize basic results of U -statistics to be used later. For details, see for example, Lehmann (1999, chap. 6), Koroljuk and Borovskich (1994), and Sering (1980, chap. 5). For a symmetric function (kernel), $h(X_1, \dots, X_m) : \mathcal{R}^m \rightarrow \mathcal{R}$ with X_i 's iid, an m th-order U -statistic, U_n , with $E(U_n) = \theta = E[h(\cdot)]$, is defined as

$$U_n = \binom{n}{m}^{-1} \sum_{1 \leq i_1 < \dots < i_m \leq n} h(X_{i_1}, \dots, X_{i_m}), \quad (14)$$

with summation over all subsets of m distinct elements taken from $\{1, \dots, n\}$. The projection $h_c(x_1, \dots, x_c) = E(h(X_1, \dots, X_m) | X_1 = x_1, \dots, X_c = x_c)$ with $h_m(\cdot) = h(\cdot)$ and $\xi_c = \text{Var}(h_c(x_1, \dots, x_c))$, $c = 1, \dots, m$, gives

$$\text{Var}(U_n) = \binom{n}{m}^{-1} \sum_{c=1}^m \binom{m}{c} \binom{n-m}{m-c} \xi_c. \quad (15)$$

If $0 < \xi_1 < \infty$ and $\xi_c < \infty \forall c = 1, \dots, m$, then (Lehmann 1999, 369)

$$\frac{U_n - E(U_n)}{\sqrt{\text{Var}(U_n)}} \xrightarrow{\mathcal{D}} N(0, 1). \quad (16)$$

For two U -statistics, $U_n^{(1)}, U_n^{(2)}$, from the same sample with kernels $h_i(X_1, \dots, X_{m_i})$, projections $h_{ic}(\cdot)$, $i = 1, 2$, let $\xi_{cc} = \text{Cov}(h_{1,c}(X_1, \dots, X_c), h_{2,c}(X_1, \dots, X_c))$ if they share c elements. Then for $m_1 \leq m_2$,

$$\text{Cov}(U_n^{(1)}, U_n^{(2)}) = \binom{n}{m_1}^{-1} \sum_{c=1}^{m_1} \binom{m_2}{c} \binom{n-m_2}{m_1-c} \xi_{cc}. \quad (17)$$

As another extension, consider the two-sample U -statistic

$$U_{n_1 n_2} = \left[\binom{n_1}{m_1} \binom{n_2}{m_2} \right]^{-1} \sum_{P(i)} \sum_{P(j)} h(X_{i_1}, \dots, X_{i_{m_1}}; X_{j_1}, \dots, X_{j_{m_2}}), \quad (18)$$

where $P(i) = 1 \leq i_1 < \dots < i_{m_1} \leq n_1$ and $P(j) = 1 \leq j_1 < \dots < j_{m_2} \leq n_2$ cover all possible subsets of m_i distinct elements taken from $\{1, \dots, n_i\}$, $i = 1, 2$, and $h(\cdot)$ is symmetric in elements within each sample. The projection $h_{c_1 c_2} = E[h(\cdot) | X_1, \dots, X_{c_1}; X_1, \dots, X_{c_2}]$ gives $\xi_{c_1 c_2} = \text{Cov}[h(\cdot), h_{c_1 c_2}(\cdot)]$, with $\xi_{c_1 c_2} = 0$, $c_1 = 0, 1, \dots, m_1$, $c_2 = 0, 1, \dots, m_2$. Then (Lee 1990, 38)

$$\text{Var}(U_{n_1 n_2}) = \left[\binom{n_1}{m_1} \binom{n_2}{m_2} \right]^{-1} \sum_{c_1=0}^{m_1} \sum_{c_2=0}^{m_2} \binom{m_1}{c_1} \binom{n_1-m_1}{m_1-c_1} \binom{m_2}{c_2} \binom{n_2-m_2}{m_2-c_2} \xi_{c_1 c_2}. \quad (19)$$

Further, if $0 \leq n_i/N \leq 1$, $N = n_1 + n_2$, $0 < \xi_{01}, \xi_{10} < \infty$, and $\xi_{c_1 c_2} < \infty$, $\forall c_i = 1, \dots, m_i$, $i = 1, 2$, then (see Lehmann 1999, Theorem 6.1.4(ii), 375)

$$\frac{U_{n_1 n_2} - E(U_{n_1 n_2})}{\sqrt{\text{Var}(U_{n_1 n_2})}} \xrightarrow{\mathcal{D}} N(0, 1). \quad (20)$$

A.2. Quadratic and bilinear forms

Theorem A.2.1 collects basic moments of quadratic and bilinear forms from the literature. In Lemma A.2.1, we extend these results to higher moments. The proofs of these results, not directly relevant here, are skipped; see Ahmad (2014).

Theorem A.2.1 (Searle 1971; Mathai et al. 1995). For u, v iid $\mathcal{N}_p(0, \Sigma)$ vectors, $\mathbf{A}_p, \mathbf{B}_p$ symmetric matrices, let $Q_1 = u' A u$, $Q_2 = u' B u$, $B = u' A v$. Then $E(Q_1) = \text{tr}(\mathbf{A}\Sigma)$, $\text{Var}(Q_1) = 2\text{tr}(\mathbf{A}\Sigma)^2$, $E(B) = 0$, $E(B^2) = \text{tr}(\mathbf{A}\Sigma)^2$, $E(B^3) = 0$, $E(B^4) = 6\text{tr}(\mathbf{A}\Sigma)^2 + 3[\text{tr}(\mathbf{A}\Sigma)]^2$, $\text{Cov}(Q_1, B) = 0$, $\text{Cov}(Q_1, Q_2) = 2\text{tr}(\mathbf{A}\Sigma \mathbf{B}\Sigma)$, $\text{Cov}(Q_1^2, Q_2) = 4\text{tr}(\mathbf{A}\Sigma)\text{tr}(\mathbf{A}\Sigma \mathbf{B}\Sigma) + 8\text{tr}[(\mathbf{A}\Sigma)^2 \mathbf{B}\Sigma]$ and

$$\begin{aligned} \text{Var}(Q_1 Q_2) &= 32\text{tr}[(\mathbf{A}\Sigma)^2 (\mathbf{B}\Sigma)^2] + 2[\text{tr}(\mathbf{A}\Sigma)]^2 \text{tr}(\mathbf{B}\Sigma)^2 + [\text{tr}(\mathbf{B}\Sigma)]^2 \text{tr}(\mathbf{A}\Sigma)^2 \\ &\quad + 4[\text{tr}(\mathbf{A}\Sigma)^2 \text{tr}(\mathbf{B}\Sigma)^2 + \text{tr}(\mathbf{A}\Sigma \mathbf{B}\Sigma)]^2 + \text{tr}(\mathbf{A}\Sigma) \text{tr}(\mathbf{B}\Sigma) \text{tr}(\mathbf{A}\Sigma \mathbf{B}\Sigma) \\ &\quad + 16[\text{tr}(\mathbf{A}\Sigma \mathbf{B}\Sigma)^2 \text{tr}(\mathbf{A}\Sigma) \text{tr}\{(\mathbf{A}\Sigma)(\mathbf{B}\Sigma)^2\} + \text{tr}(\mathbf{B}\Sigma) \text{tr}\{(\mathbf{A}\Sigma)^2 (\mathbf{B}\Sigma)\}] \end{aligned}$$

For Lemma A.2.1, and subsequent proofs, let $\mathbf{a}_{it} = \mathbf{X}_{it} - \boldsymbol{\mu}_i$ with $\mathbf{a}_{it} \sim \mathcal{N}(\mathbf{0}, \Sigma_i)$, where $\mathbf{a}_{it}$ and $\mathbf{a}_{it'}$ are independent for $t \neq t'$.

Lemma A.2.1. Let $\mathbf{a}_{it}$ be as already defined. Then

$$E[\mathbf{a}'_{it} \mathbf{a}_{iu} \mathbf{a}'_{it} \mathbf{a}_{iv} \mathbf{a}'_{iu} \mathbf{a}_{iv}] = \text{tr}(\mathbf{a}_i^4) \quad (21)$$

$$E[\mathbf{a}'_{it} \mathbf{a}_{iu} \mathbf{a}'_{iw} \mathbf{a}_{iu} \mathbf{a}'_{it} \mathbf{a}_{iv} \mathbf{a}'_{iw} \mathbf{a}_{iv}] = \text{tr}(\mathbf{a}_i^4) \quad (22)$$

$$E[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2\mathbf{a}'_{it}\mathbf{a}_i\mathbf{a}_{it}] = 2\text{tr}(\mathbf{a}_i^4) + [\text{tr}(\mathbf{a}_i^2)]^2 \quad (23)$$

$$\text{Var}(\mathbf{a}'_{it}\mathbf{a}_{iu}\mathbf{a}'_{iv}\mathbf{a}_{iu}) = 2\text{tr}(\mathbf{a}_i^4) + [\text{tr}(\mathbf{a}_i^2)]^2 \quad (24)$$

$$\text{Cov}[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2, (\mathbf{a}'_{it}\mathbf{a}_{iv})^2] = 2\text{tr}(\mathbf{a}_i^4) \quad (25)$$

$$E[(\mathbf{a}'_{it}\mathbf{a}_{ju})^2\mathbf{a}'_{it}\mathbf{a}_j\mathbf{a}_{it}] = 2\text{tr}\{(\mathbf{a}_j\mathbf{a}_j)^2\} + [\text{tr}(\mathbf{a}_j\mathbf{a}_j)]^2 \quad (26)$$

$$\text{Var}(\mathbf{a}'_{it}\mathbf{a}_{ju}\mathbf{a}'_{iv}\mathbf{a}_{ju}) = 2\text{tr}\{(\mathbf{a}_j\mathbf{a}_j)^2\} + [\text{tr}(\mathbf{a}_j\mathbf{a}_j)]^2 \quad (27)$$

$$\text{Cov}[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2, (\mathbf{a}'_{it}\mathbf{a}_{iv})^2] = 2\text{tr}\{(\mathbf{a}_j\mathbf{a}_j)^2\} \quad (28)$$

$$E(\mathbf{a}'_{it}\mathbf{a}_{iu}\mathbf{a}'_{jt}\mathbf{a}_{iv}\mathbf{a}'_{iu}\mathbf{a}_j\mathbf{a}_{iv}) = \text{tr}\{(\mathbf{a}_j\mathbf{a}_j)^2\} \quad (29)$$

$$\text{Cov}[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2, \mathbf{a}'_{it}\mathbf{a}_j\mathbf{a}_{it}] = 2\text{tr}(\mathbf{a}_i^3\mathbf{a}_j) \quad (30)$$

$$\text{Cov}(\mathbf{a}'_{it}\mathbf{a}_j\mathbf{a}_{iu}, \mathbf{a}'_{it}\mathbf{a}_j\mathbf{a}_{iu}) = \text{tr}\{(\mathbf{a}_i^3\mathbf{a}_j)\} \quad (31)$$

$$E[(\mathbf{a}'_{iu}\mathbf{a}_{iv})^2\mathbf{a}'_{it}\mathbf{a}_j\mathbf{a}_{it}] = \text{tr}(\mathbf{a}_i\mathbf{a}_j)\text{tr}(\mathbf{a}_i)^2, \quad (32)$$

where $E[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2\mathbf{a}'_{it}\mathbf{a}_{iu}\mathbf{a}'_{iv}\mathbf{a}_{iu}]$, $E[\mathbf{a}'_{it}\mathbf{a}_{iu}\mathbf{a}'_{it}\mathbf{a}_{iv}\mathbf{a}'_{it}\mathbf{a}_j\mathbf{a}_{it}]$, $E[\mathbf{a}'_{it}\mathbf{a}_{iu}\mathbf{a}'_{it}\mathbf{a}_{iv}\mathbf{a}'_{it}\mathbf{a}_j\mathbf{a}_{iu}]$, $E[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2\mathbf{a}'_{it}\mathbf{a}_j\mathbf{a}_{iu}]$, $E[(\mathbf{a}'_{it}\mathbf{a}_{iu})^2\mathbf{a}'_{it}\mathbf{a}_j\mathbf{a}_{iu}]$, and $E[\mathbf{a}'_{it}\mathbf{a}_{iu}\mathbf{a}'_{it}\mathbf{a}_{iv}\mathbf{a}'_{it}\mathbf{a}_{iv}\mathbf{a}'_{it}\mathbf{a}_{iu}]$ all vanish.

Appendix B: Main proofs

B.1. Proof of Theorem 2.1.1

As the computations are quite involved, we only provide essential results; for details, see Ahmad (2014). Consider $E_1 = U_{n_1} = U/12$ as a U -statistic with (section A.1) $h(\cdot) = A_{1krk'r'}^2 + A_{1kk'r'r'}^2 + A_{1kr'k'r}^2 = (\mathbf{D}'_{1kr}\mathbf{D}_{1k'r'})^2 + (\mathbf{D}'_{1kk'}\mathbf{D}_{1rr'})^2 + (\mathbf{D}'_{1kr'}\mathbf{D}_{1k'r})^2$, ignoring 12 for convenience. Then, with a_{it} as already defined,

$$\begin{aligned} h(\cdot) &= (\mathbf{a}'_{1k}\mathbf{a}_{1k'} - \mathbf{a}_{1k}\mathbf{a}_{1r'} - \mathbf{a}'_{1r}\mathbf{a}_{1k'} + \mathbf{a}'_{1r}\mathbf{a}_{1r'})^2 \\ &\quad + (\mathbf{a}'_{1k}\mathbf{a}_{1r} - \mathbf{a}_{1k}\mathbf{a}_{1r'} - \mathbf{a}'_{1k'}\mathbf{a}_{1r} + \mathbf{a}'_{1k'}\mathbf{a}_{1r'})^2 \\ &\quad + (\mathbf{a}'_{1k}\mathbf{a}_{1k'} - \mathbf{a}_{1k}\mathbf{a}_{1r} - \mathbf{a}'_{1r'}\mathbf{a}_{1k'} + \mathbf{a}'_{1r'}\mathbf{a}_{1r})^2 \\ &= 2(\mathbf{a}'_{1k}\mathbf{a}_{1k'})^2 + 2(\mathbf{a}'_{1r}\mathbf{a}_{1k'})^2 + 2(\mathbf{a}'_{1r}\mathbf{a}_{1r'})^2 + 2(\mathbf{a}'_{1k'}\mathbf{a}_{1k'})^2 \\ &\quad - 2\mathbf{a}'_{1k'}\mathbf{a}_{1k}\mathbf{a}'_{1k'}\mathbf{a}_{1r'} - 2\mathbf{a}'_{1k}\mathbf{a}_{1k'}\mathbf{a}'_{1k'}\mathbf{a}_{1r} + 4\mathbf{a}'_{1k}\mathbf{a}_{1k'}\mathbf{a}'_{1r}\mathbf{a}_{1r'} \\ &\quad + 4\mathbf{a}'_{1k}\mathbf{a}_{1r'}\mathbf{a}'_{1r}\mathbf{a}_{1k'} - 2\mathbf{a}'_{1k}\mathbf{a}_{1r'}\mathbf{a}'_{1r'}\mathbf{a}_{1r} - 2\mathbf{a}'_{1k'}\mathbf{a}_{1r}\mathbf{a}'_{1r}\mathbf{a}_{1r'} \\ &\quad - 2\mathbf{a}'_{1r}\mathbf{a}_{1k}\mathbf{a}'_{1k'}\mathbf{a}_{1r'} - 2\mathbf{a}'_{1k}\mathbf{a}_{1r}\mathbf{a}'_{1r}\mathbf{a}_{1k'} - 2\mathbf{a}'_{1k}\mathbf{a}_{1r'}\mathbf{a}'_{1r'}\mathbf{a}_{1k'} \\ &\quad - 2\mathbf{a}'_{1k'}\mathbf{a}_{1k}\mathbf{a}'_{1k'}\mathbf{a}_{1r} - 2\mathbf{a}'_{1k}\mathbf{a}_{1k'}\mathbf{a}'_{1k'}\mathbf{a}_{1r'} - 2\mathbf{a}'_{1k}\mathbf{a}_{1r}\mathbf{a}'_{1r}\mathbf{a}_{1r'} - 2\mathbf{a}'_{1k'}\mathbf{a}_{1r'}\mathbf{a}'_{1r'}\mathbf{a}_{1r} \end{aligned}$$

so that $h_c(\cdot) = E[h(\cdot)|\mathbf{a}_{1k}, \dots, \mathbf{a}_{1r'}]$ and $\xi_{cc} = \text{Var}[h_c(\cdot)]$ for $c = 1, 2, 3$ are:

$$\begin{aligned}
h_1(\cdot) &= 6\mathbf{a}'_{1k}\mathbf{a}_1\mathbf{a}_{1k} + 6\text{tr}(\mathbf{a}_1^2)\text{arrow}\xi_1 = 36[2\text{tr}(\mathbf{a}_1^4)] = 72\text{tr}(\mathbf{a}_1^4) \\
h_2(\cdot) &= 4\mathbf{a}'_{1k}\mathbf{a}_1\mathbf{a}_{1k} + 4\mathbf{a}'_{1r}\mathbf{a}_1\mathbf{a}_{1r} + 2(\mathbf{a}'_{1k}\mathbf{a}_{1r})^2 - 4\mathbf{a}'_{1k}\mathbf{a}_1\mathbf{a}_{1r} + 2\text{tr}(\mathbf{a}_1^2) \\
\xi_2 &= 168\text{tr}(\mathbf{a}_1^4) + 8[\text{tr}(\mathbf{a}_1^2)]^2 \\
h_3(\cdot) &= 2[(\mathbf{a}'_{1k}\mathbf{a}_{1k'})^2 + \mathbf{a}'_{1k}\mathbf{a}_1\mathbf{a}_{1k} + (\mathbf{a}'_{1r}\mathbf{a}_{1k'})^2 + \mathbf{a}'_{1r}\mathbf{a}_1\mathbf{a}_{1r} \\
&\quad - \mathbf{a}'_{1k}\mathbf{a}_{1k'}\mathbf{a}'_{1r}\mathbf{a}_{1k'} - \mathbf{a}'_{1k}\mathbf{a}_{1r}\mathbf{a}'_{1k'}\mathbf{a}_{1r} - \mathbf{a}'_{1k}\mathbf{a}_{1k'}\mathbf{a}'_{1k}\mathbf{a}_{1r} \\
&\quad - \mathbf{a}'_{1k}\mathbf{a}_1\mathbf{a}_{1r} + (\mathbf{a}'_{1k}\mathbf{a}_{1r})^2 + \mathbf{a}'_{1k'}\mathbf{a}_1\mathbf{a}_{1k'} - \mathbf{a}'_{1k'}\mathbf{a}_1\mathbf{a}_{1k'} - \mathbf{a}'_{1k'}\mathbf{a}_1\mathbf{a}_{1r}] \\
\xi_3 &= 300\text{tr}(\mathbf{a}_1^4) + 36[\text{tr}(\mathbf{a}_1^2)]^2.
\end{aligned}$$

For $c = 4$ with $h_4(\cdot) = h(\cdot)$, we have

$$\begin{aligned}
\xi_4 &= \text{Var}[(\mathbf{D}'_{1kr}\mathbf{D}_{1k'r'})^2 + (\mathbf{D}'_{1kk'}\mathbf{D}_{1rr'})^2 + (\mathbf{D}'_{1kr'}\mathbf{D}_{1k'r})^2] \\
&= 3\text{Var}[(\mathbf{D}'_{1kr}\mathbf{D}_{1k'r'})^2] + 3 \cdot 2\text{Cov}[(\mathbf{D}'_{1kr}\mathbf{D}_{1k'r'})^2, (\mathbf{D}'_{1kk'}\mathbf{D}_{1rr'})^2]
\end{aligned}$$

with $\text{Var}[(\mathbf{D}'_{1kr}\mathbf{D}_{1k'r'})^2]$ from Theorem A.2.1 and for $\text{Cov}[(\mathbf{D}'_{1kr}\mathbf{D}_{1k'r'})^2, (\mathbf{D}'_{1kk'}\mathbf{D}_{1rr'})^2]$, $E[(\mathbf{D}'_{1kr}\mathbf{D}_{1k'r'})^2(\mathbf{D}'_{1kk'}\mathbf{D}_{1rr'})^2] = E(A_{1krk'r'}^2 A_{1kk'rr'}^2)$ simplifies to

$$E(A_{1krk'r'}^2 A_{1kk'rr'}^2) = E[\{(\mathbf{a}_{1k} - \mathbf{a}_{1r})(\mathbf{a}_{1k'} - \mathbf{a}_{1r'})\}'\{(\mathbf{a}_{1k} - \mathbf{a}_{1k}')(\mathbf{a}_{1r} - \mathbf{a}_{1r'})\}]^2,$$

as the joint product moment of two bilinear forms each of independent components. Let

$$\mathbf{u} = (\mathbf{a}_{1k} - \mathbf{a}_{1r}\mathbf{a}_{1k'} - \mathbf{a}_{1r'})', \quad \mathbf{v} = (\mathbf{a}_{1k} - \mathbf{a}_{1k'}\mathbf{a}_{1r} - \mathbf{a}_{1r'})' \quad \text{with} \quad \text{Cov}(\mathbf{u}) = \begin{pmatrix} \boldsymbol{\Sigma}_i & \mathbf{O} \\ \mathbf{O} & \boldsymbol{\Sigma}_i \end{pmatrix} =$$

$\Lambda = \text{Cov}(\mathbf{v})$. With $\mathbf{A} = \begin{pmatrix} \mathbf{O} & \mathbf{I} \\ \mathbf{I} & \mathbf{O} \end{pmatrix}$, $\text{tr}(\mathbf{A}\Lambda) = 0$, $\text{tr}(\mathbf{A}\Lambda)^2 = 4 \text{tr}(\boldsymbol{\Sigma}_1^2)$, and $\text{tr}(\mathbf{A}\Lambda)^4 = 16\text{tr}(\boldsymbol{\Sigma}_1^4)$. With $\mathbf{Q}_1 = \mathbf{u}'\mathbf{A}\mathbf{u}$, $\mathbf{Q}_2 = \mathbf{v}'\mathbf{A}\mathbf{v}$, $E(A_{1krk'r'}^2 A_{1kk'rr'}^2) = E(\mathbf{Q}_1^2 \mathbf{Q}_2^2)/16 = 96 \text{tr}(\boldsymbol{\Sigma}_1^4) + 48[\text{tr}(\boldsymbol{\Sigma}_1^2)]^2$, $\text{Cov}[(\mathbf{D}'_{1kr}\mathbf{D}_{1k'r'})^2, (\mathbf{D}'_{1kk'}\mathbf{D}_{1rr'})^2] = 96 \text{tr}(\boldsymbol{\Sigma}_1^4) + 32[\text{tr}(\boldsymbol{\Sigma}_1^2)]^2$, and $\xi_4 = 96\{9 \text{tr}(\boldsymbol{\Sigma}_1^4) + 3[\text{tr}(\boldsymbol{\Sigma}_1^2)]^2\}$. Dividing all ξ_i 's by 144 (for the divisor 12), substituting the results in Eq. (19), and simplifying yields the required result.

For $\text{Var}(E_{12})$, $E_{12} = U_{n_1 n_2} = U/4$ and $h(\cdot) = A_{12krls}^2 = (\mathbf{D}'_{1kr}\mathbf{D}_{2ls})^2$ simplifies to $[(\mathbf{a}_{1k} - \mathbf{a}_{1r})'(\mathbf{a}_{2l} - \mathbf{a}_{2s})]^2 = [\mathbf{a}'_{1k}\mathbf{a}_{2l} - \mathbf{a}'_{1k}\mathbf{a}_{2s} - \mathbf{a}'_{1r}\mathbf{a}_{2l} + \mathbf{a}'_{1r}\mathbf{a}_{2s}]^2$. With $h_{c_1 c_2}(\cdot) = E[h(\cdot)|\mathbf{x}_{1k}, \mathbf{x}_{2l}, \dots]$, $\xi_{c_1 c_2} = \text{Var}(h_{c_1 c_2})$, $c_i = 0, 1, 2$, $i = 1, 2$, we get

$$\begin{aligned}
h_{10} &= 2\mathbf{a}'_{1k}\boldsymbol{\Sigma}_2\mathbf{a}_{1k} + 2 \text{tr}(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2) \Rightarrow \xi_{10} = 8 \text{tr}\{(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2)^2\} \\
h_{20} &= 2(\mathbf{a}_{1k} - \mathbf{a}_{1r})'\boldsymbol{\Sigma}_2(\mathbf{a}'_{1k} - \mathbf{a}_{1r}) \Rightarrow \xi_{20} = 32\{\text{tr}(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2)^2\} \\
h_{11} &= (\mathbf{a}'_{1k}\mathbf{a}_{2l})^2 + \mathbf{a}'_{1k}\boldsymbol{\Sigma}_2\mathbf{a}_{1k} + \mathbf{a}'_{2l}\boldsymbol{\Sigma}_1\mathbf{a}_{2l} + \text{tr}(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2) \\
\xi_{11} &= 18 \text{tr}\{(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2)^2\} + 2[\text{tr}(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2)]^2 \\
h_{21} &= (\mathbf{a}_{1k} - \mathbf{a}_{1r})'\boldsymbol{\Sigma}_2(\mathbf{a}_{1k} - \mathbf{a}_{1r}) + [(\mathbf{a}_{1k} - \mathbf{a}_{1r})'\mathbf{a}_{2l}]^2 \\
\xi_{21} &= 48 \text{tr}\{(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2)^2\} + 8[\text{tr}(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2)]^2 \\
h_{22} &= h(\cdot) \Rightarrow \xi_{22} = \text{Var}(A_{12krls}^2) = 96 \text{tr}\{(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2)^2\} + 32[\text{tr}(\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_2)]^2
\end{aligned}$$

with $\xi_{10} = \xi_{01}$, $\xi_{20} = \xi_{02}$ and so on, by symmetry. Dividing all ξ 's by 16 (to count for divisor 4), substituting in Eq. (19) and simplifying gives $\text{Var}(E_{12})$. Now, consider the covariances. For $\text{Cov}(E_1, E_{12})$, first $E[D_{1krk'r'}^2 A_{12klrs}^2]$ simplifies to $E[(\mathbf{D}'_{1kr}\boldsymbol{\Sigma}_2\mathbf{D}_{1kr})\mathbf{D}_{1krk'r'}^2]$, using trace properties, so that $E(E_1, E_{12})$ reduces to

$$\begin{aligned}
E(E_1 E_{12}) &= \frac{1}{48P(n_1)P(n_1, n_2)} \sum_{k=1}^{n_1} \sum_{\substack{r=1 \\ k \neq r}}^{n_1} \sum_{l=1}^{n_2} \sum_{\substack{s=1 \\ l \neq s}}^{n_2} \sum_{k=1}^{n_1} \sum_{\substack{r=1 \\ \pi(k, r, k', r')}}^{n_1} \sum_{k'=1}^{n_1} \sum_{r'=1}^{n_1} E[A_{12krls}^2 D_{1krk'r'}^2] \\
&= \frac{1}{24P(n_1)Q(n_1)} \sum_{k=1}^{n_1} \sum_{\substack{r=1 \\ k \neq r}}^{n_1} \sum_{k=1}^{n_1} \sum_{\substack{r=1 \\ \pi(k, r, k', r')}}^{n_1} \sum_{k'=1}^{n_1} \sum_{r'=1}^{n_1} E[(D'_{kr} \Sigma_2 D_{kr}) D_{1krk'r'}^2] \\
&= E \left[\frac{1}{2Q(n_1)} \sum_{k=1}^{n_1} \sum_{\substack{r=1 \\ k \neq r}}^{n_1} (D'_{1kr} \Sigma_2 D_{1kr}) \cdot \frac{1}{12P(n_1)} \sum_{k=1}^{n_1} \sum_{r=1}^{n_1} \sum_{k'=1}^{n_1} \sum_{r'=1}^{n_1} D_{1krk'r'}^2 \right] \\
&= E(U_1 U_2) \Rightarrow \text{Cov}(E_1, E_{12}) = \text{Cov}(U_1, U_2)
\end{aligned}$$

which follows from Eq. (17) with $m_1 = 2$, $m_2 = 4$, and ξ_{11} , ξ_{22} . Writing $\mathbf{D}'_{1kr} \mathbf{a}_2 \mathbf{D}_{1kr} = \mathbf{a}'_{1k} \mathbf{a}_2 \mathbf{a}_{1k} - 2\mathbf{a}'_{1k} \mathbf{a}_2 \mathbf{a}_{1r} + \mathbf{a}'_{1r} \mathbf{a}_2 \mathbf{a}_{1r}$, we have

$$\begin{aligned}
h_{11} &= E[h(\mathbf{X}_{1k}, \mathbf{X}_{1r}) | \mathbf{x}_{1k}] = \mathbf{a}'_{1k} \Sigma_2 \mathbf{a}_{1k} + \text{tr}(\Sigma_1 \Sigma_2) \\
h_{12} &= E[h(\mathbf{X}_{1k}, \mathbf{X}_{1r}) | \mathbf{x}_{1k}, \mathbf{x}_{1r}] = \mathbf{D}'_{1kr} \Sigma_2 \mathbf{D}_{1kr} = h_1(\cdot) \\
h_{21} &= 6\mathbf{a}'_{1k} \Sigma_1 \mathbf{a}_{1k} + 6 \text{tr}(\Sigma_1^2) \\
h_{22} &= 4\mathbf{a}'_{1k} \Sigma_1 \mathbf{a}_{1k} + 4\mathbf{a}'_{1r} \Sigma_1 \mathbf{a}_{1r} + 2(\mathbf{a}'_{1k} \mathbf{a}_{1r})^2 - 4\mathbf{a}'_{14} \Sigma_1 \mathbf{a}_{1r} + 2\text{tr}(\Sigma_1^2),
\end{aligned}$$

so that $\xi_{11} = \text{Cov}(h_{11}, h_{21}) = 12 \text{tr}(\Sigma_1^3 \Sigma_2)$, and $\xi_{22} = \text{Cov}(h_{12}, h_{22}) = 32 \text{tr}(\Sigma_1^3 \Sigma_2)$. Substitution gives the required result. Finally, the bounds follow by Cauchy-Schwarz inequality using, for example, $\text{tr}(\Sigma_1^4) \leq [\text{tr}(\Sigma_1^2)]^2$.

B.2. Proof of Theorem 2.1.2

For $\tilde{E}_1 = E_1 / \text{tr}(\Sigma_1^2) - 1 = \tilde{U}_{n_1}$ with $\tilde{h} = h(\cdot) / \text{tr}(\Sigma_1^2) - 1$, as defined after Eq. (4), let the first-order projection of $\tilde{U}_n$ be $\hat{U}_{n_1} = m \sum_{k=1}^{n_1} g(\mathbf{x}_{1k}) / n_1$ where (vander Vaart 1998, 163; Serfling 1980, 178)

$$g(\mathbf{x}_{1k}) = \tilde{h}_1(\mathbf{x}_{1k}) = \frac{h_1}{\text{tr}(\Sigma_1^2)} - 1 = \frac{1}{\text{tr}(\Sigma_1^2)} \cdot \frac{1}{2} [\mathbf{a}'_{1k} \Sigma_1 \mathbf{a}_{1k} + \text{tr}(\Sigma_1^2)] - 1.$$

With $h_1(\cdot)$, ξ_1 from section B.1, $E[g(\mathbf{x}_{1k})] = 0$ and $\text{Var}[g(\mathbf{x}_{1k})] = \text{tr}(\Sigma_1^4) / 2 [\text{tr}(\Sigma_1^2)]^2$. This gives $E(\hat{U}_{n_1}) = 0$ and $\text{Var}(\hat{U}_{n_1}) = 8 \text{tr}(\Sigma_1^4) / n_1 [\text{tr}(\Sigma_1^2)]^2 \leq 8 / n_1 = O(1/n_1)$, where the bound is independent of p . Now, consider $\text{Cov}(\tilde{U}_{n_1}, \hat{U}_{n_1})$, which, with $m_1 = 1$, $m_2 = 4$, follows from Eq. (17) by only computing ξ_{11} . With kernels, $g_1(\cdot)$, $h_1(\cdot)$, ξ_{11} reduces to $\text{Var}(\mathbf{a}'_{1k} \Sigma_1 \mathbf{a}_{1k}) = 2\mathbf{a}'_{1k} \Sigma_1 \mathbf{a}_{1k} + 6 \text{tr}(\Sigma_1^4)$, so that $\text{Cov}(\tilde{U}_{n_1}, \hat{U}_{n_1}) = \text{Var}(\hat{U}_{n_1})$, as required.

Hence, $\tilde{U}_{n_1} = \hat{U}_{n_1} + o_p(1)$ with $\text{Var}(\tilde{U}_{n_1}) / \text{Var}(\hat{U}_{n_1}) \rightarrow 1$. Since, $\hat{U}_{n_1}$, being a sum of iid random variables, follows asymptotic normality by the usual CLT, the normality of $\tilde{U}_{n_1}$ follows from Slutsky's lemma (see van der Vaart 1998, chap. 12). Since the kernel of U_{n_1} is of order 4, we need higher order projections with similar arguments. For example, for $\tilde{h}_2(\mathbf{x}_{1k}, \mathbf{x}_{1r})$, $g_2(\mathbf{x}_{1k}, \mathbf{x}_{1r}) = \tilde{h}_2(\mathbf{x}_{1k}, \mathbf{x}_{1r}) - g(\mathbf{x}_{1k}) - g(\mathbf{x}_{1r})$ (see Serfling 1980, 178), which simplifies to

$$g_2(\cdot) = \frac{-1}{6[\text{tr}(\mathbf{\Sigma}_1^2)]^2} [(\mathbf{a}_{1k} + \mathbf{a}_{1r})\mathbf{\Sigma}(\mathbf{a}_{1k} - \mathbf{a}_{1r}) + (\mathbf{a}'_{1k}\mathbf{a}_{1r})^2 - \text{tr}(\mathbf{\Sigma}_1^2)],$$

with $E[g_2(\cdot)] = 0$ and $\text{Var}[g_2(\cdot)] = [3\text{tr}(\mathbf{\Sigma}_1^4) + [\text{tr}(\mathbf{\Sigma}_1^2)]^2]/18[\text{tr}(\mathbf{\Sigma}_1^2)]^2$. The second-order projection is (van der Vaart 1998, chap. 12; Serfling 1980, chap. 6)

$$\hat{U}_{n_1} = \frac{m(m-1)}{Q(n_1)} \sum_{k=1}^{n_1} \sum_{\substack{r=1 \\ k \neq r}}^{n_1} g_2(\mathbf{x}_{1k}, \mathbf{x}_{1r}) = \frac{12}{Q(n_1)} \sum_{k=1}^{n_1} \sum_{\substack{r=1 \\ k \neq r}}^{n_1} g_2(\mathbf{x}_{1k}, \mathbf{x}_{1r}),$$

with $E(\hat{U}_{n_1}) = 0$, $\text{Var}(\hat{U}_{n_1}) = 8[3\text{tr}(\mathbf{\Sigma}_1^4) + [\text{tr}(\mathbf{\Sigma}_1^2)]^2]/Q(n_1)\text{tr}(\mathbf{\Sigma}_1^2)]^2 = O(1/n_1^2)$, which is uniformly bounded in p , where $Q(n_1) = n_1(n_1 - 1)$. It can again be verified that $\text{Cov}(\tilde{U}_{n_1}, \hat{U}_{n_1}) = \text{Var}(\hat{U}_{n_1})$, so that the result follows similarly as for first-order projection. As $\text{Var}(\hat{U}_{n_1})$ is in increasing order of $1/n_1$ with increasing c , the final limit follows from Eq. (16) in section A.1.

The proof for $\tilde{E}_{12}$ follows similarly by using first-order projection from both components, exactly the same way as for $\tilde{E}_1$. With $\tilde{h}(\cdot) = A_{12krls}^2/\text{tr}(\mathbf{\Sigma}_1\mathbf{\Sigma}_2) - 1$ and $g_{10} = \tilde{h}_{10}$, $g_{01} = \tilde{h}_{01}$, the projection is given as (van der Vaart 1998, 165)

$$\begin{aligned} \hat{U}_{n_1n_2} &= \frac{m_1}{n_1} \sum_{k=1}^{n_1} g_{10}(\mathbf{x}_{1k}) + \frac{m_2}{n_2} \sum_{l=1}^{n_2} g_{01}(\mathbf{x}_{2l}) \\ &= \frac{1}{\text{tr}(\mathbf{\Sigma}_1\mathbf{\Sigma}_2)} \left[\frac{1}{n_1} \sum_{k=1}^{n_1} \mathbf{a}'_{1k}\mathbf{\Sigma}_2\mathbf{a}_{1k} + \frac{1}{n_2} \sum_{l=1}^{n_2} \mathbf{a}'_{2l}\mathbf{\Sigma}_1\mathbf{a}_{2l} - 2\text{tr}(\mathbf{\Sigma}_1\mathbf{\Sigma}_2) \right] \end{aligned}$$

with $E(\hat{U}_{n_1n_2}) = 0$ and $\text{Var}(\hat{U}_{n_1n_2}) = 2(1/n_1 + 1/n_2)\text{tr}(\mathbf{\Sigma}_1\mathbf{\Sigma}_2)^2/[\text{tr}(\mathbf{\Sigma}_1\mathbf{\Sigma}_2)]^2$, independent of p . With the components independent, the proof follows by treating each component as in the one-sample case and using Eq. (20).